# The Paleomagnetism of the Otish Gabbro from North of the Grenville Front, Quebec

W. F. FAHRIG

*Geological Survey of Canada, Ottawa, Canada*

AND

E. H. CHOWN

*Loyola College, Montreal, Quebec*

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The Otish Gabbro, which intrudes the Otish Mountains Group, is normally magnetized and has a paleomagnetic pole at 107° W, 35° N, $\alpha_{95} = 9°$. The Otish Mountains Group has been correlated with the Mistassini Group and/or the Huronian Supergroup. The pole for the Otish Gabbro lies between that of the 2175 m.y. old Nipissing diabase, which is intrusive into Huronian strata, and the pole for the 1700 m.y. old Sudbury norite. If the Otish Mountains Group is correlative with the 2300 m.y. Huronian, the Otish Gabbro was intruded a considerable time after the deposition of the sedimentary strata.

Le gabbro Otish qui injecte le groupe des Monts Otish est habituellement magnétisé et son pôle paléomagnétique est 107° W, 35° N, $\alpha_{95} = 9°$. Le groupe des Monts Otish correspond au groupe de Mistassini et/ou au supergroupe Huronien. Dans le cas du gabbro Otish, le pôle est entre celui du vieux diabase de Nipissing (2175 m.a.) qui injecte les strates huroniennes et celui de la norite de Sudbury (1700 m.a.). Si le groupe des Monts Otish correspond à l'Huronien de 2300 m.a., le gabbro Otish a été injecté longtemps après le dépôt des strates sédimentaires.

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## Introduction

Rocks of the Superior Orogen have not been significantly affected by either thermal or deformation events since the end of the Kenoran orogeny about 2500 m.y. ago. For this reason it has been suggested (Roy and Fahrig 1973) that this vast province be considered the reference plate in paleomagnetic studies of Precambrian rocks of North America. Rocks from within the Superior, (or those whose position relative to the Superior Orogen since their formation can be determined from independent evidence) are of particular importance: they provide key points for the Precambrian North American polar wandering path.

Some of the Otish Mountains Group (Bergeron 1957), with its sills of Otish Gabbro, lies essentially undisturbed upon the eastern fringe of the Superior Orogen. Because of this, the Otish Gabbro was chosen as a subject for paleomagnetic study. Sampling was restricted to the gently south-dipping northern limb of the Otish syncline, where strata are not believed to have been affected by major movements.

## Geology

The Otish Gabbros comprise three major sills with minor subsidiary sills and feeder dikes (Chown 1970). The intrusions extend for 105 km (65 mi) along the strike of, and 80 km (50 mi) across the strike of, the Otish sedimentary basin. For ease of reference three main sills have been named; the Novet (lowest), Margat (middle), and Conflans (highest), after the lakes near which they are well exposed. The exposures near these lakes constitute the type areas of the three major intrusions (Fig. 1). The sills generally conform to bedding, but change stratigraphic levels along steeply inclined sheets and dikes, appearing progressively higher in the section to the east.

The Novet Sill intrudes the sedimentary rocks near the base of the section and extends across into the basement (Chown 1968; 1970). It attains a thickness of at least 120 m (400 ft) and is composed largely of fresh black olivine gabbro. This sill was not sampled in the present study; however, the dike sampled at site 61 (Fig. 1) is thought to be a feeder to the Novet.

The Margat Sill attains a thickness of at least 250 m (800 ft). It also intrudes basement gneisses, but moves higher in the section and is replaced at the unconformity by the

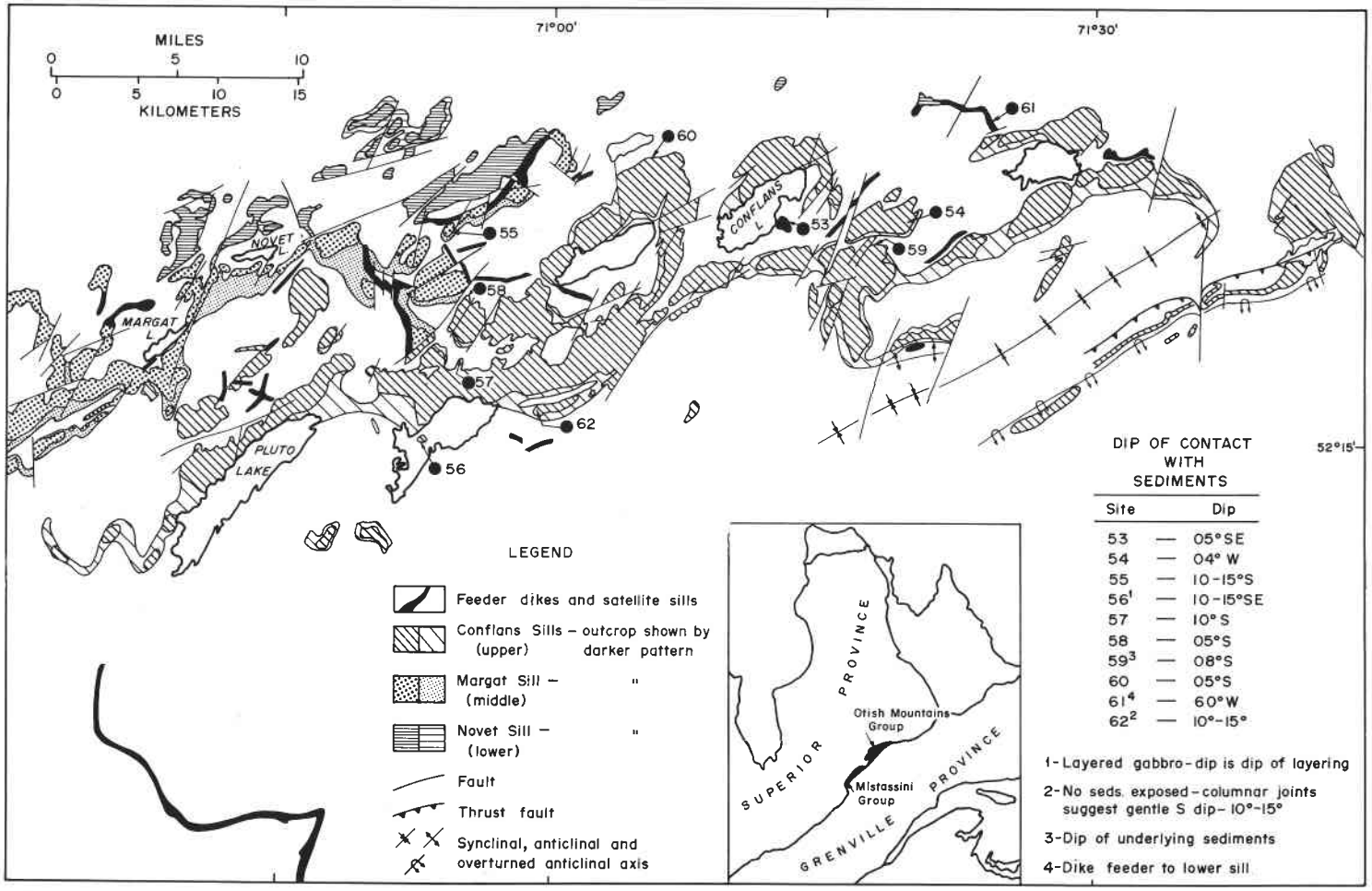



FIG. 1. Map showing the distribution of the major Otish Gabbro intrusions and the location of sampling sites on the north flank of the Otish Mountains. Geology compiled from Chown (1968, 1969, 1970).

TABLE 1. Chemical composition of hypabyssal Otish Gabbro and Seal diabase intrusions

| | $SiO_2$ | $Al_2O_3$ | $Fe_2O_3$ | FeO | MgO | CaO | $Na_2O$ | $K_2O$ | $TiO_2$ | MnO | $P_2O_5$ | $FeO/Fe_2O_3$ |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Otish Gabbro[1] | 48.1 | 16.0 | 3.4 | 9.9 | 6.7 | 8.9 | 3.2 | 1.1 | 2.1 | 0.21 | 0.29 | 2.9 |
| Seal diabase[2] | 46.7 | 16.1 | 3.8 | 9.4 | 6.4 | 9.0 | 3.2 | 0.7 | 2.2 | 0.19 | 0.28 | 3.0 |

[1]Mean composition of 5 fine-grained specimens of Otish Gabbro. Samples 44U3, 44U4, 24Z2, 23Z1, 24Z3. Analyst: Z. Katzendorfer, Quebec Dep. Natur. Resour.
[2]Mean composition of 10 Seal diabase specimens (Roy and Fahrig 1973).

Novet Sill. The Margat and Novet Sills converge on a dike northeast of sample site 55 (Fig. 1).

The uppermost Conflans Sill is up to 300 m (1000 ft) thick and is the most widespread. This sill is noticeably differentiated, and is layered in some localities (sample site 56). Both the Margat and Conflans Sills have been intensely uralitized—an alteration resulting in the conversion of the original pyroxene and some of the olivine to pale green amphibole. Olivine is more commonly converted to serpentine or talc with webs of fine-grained magnetite. Uralitized gabbros normally contain small crystals of pyrite formed at the expense of magnetite. Large scale replacement of olivine and pyroxene by chlorite is locally observed in the sills and is especially common in the smaller dikes.

Table 1 gives the mean composition of five fine-grained samples of Otish Gabbro, recalculated on a water-free basis. The five samples differ only slightly in composition, and are similar in composition to the hypabyssal Seal diabases (Roy and Fahrig 1973). The Otish rocks appear to be significantly higher in potash. Although the Otish Gabbro has undergone extensive deuteric alteration, the ferrous–ferric ratios of the two sets of intrusions are very similar. The alkali–silica ratios of the Otish rocks places them in the alkali basalt field, as defined by MacDonald and Katsura (1964).

## Age of the Otish Gabbro

The age of the Otish Gabbro is not precisely known. One specimen of the upper sill from near Conflans Lake yielded a whole-rock K–Ar age of 1465 m.y. (Wanless *et al.* 1965). Chown (1970) has argued that the uralitization and chloritization of the Otish Gabbro was related to cooling of the gabbros in the presence of abundant intrastratal water provided by the sediments of the Otish Mountains Group. He observed, for example, that the lower sill is least altered and this sill is largely restricted to the dry basement rocks. This may suggest that the intrusion of the gabbro took place early in the development of the basin, and prior to partial cementation of the clastic rocks.

The Otish Mountains Group has been correlated with the Mistassini Group to the southwest (Bergeron 1957; Chown and Caty 1973). Frarey and Roscoe (1970) suggested that the Otish Mountains Group is correlative with the Papaskwasati Formation of the Mistassini Group, which they consider to be significantly older than the rest of the Mistassini Group. This relationship is not accepted by Chown and Caty, who note a transition between the clastic and carbonate units in the Mistassini Group and stromatolitic dolomite within the clastic units of the Otish Mountains Group.

Fryer (1972) obtained a Rb/Sr isochron age of 1787 ± 55 m.y.[1] from the Temiscamie Formation of the Mistassini. This may not be the age of sedimentation, but rather the time when diagenesis or possibly very low grade metamorphism homogenized the Sr isotopes. A spectacular northwest-trending set of gabbro dikes, which strike towards the Mistassini Group, is observed to cut only Papaskwasati Formation, although a similar mass of gabbro cuts the Albanel Formation in the center of the basin (Neilson 1966). K–Ar ages of 1960 ± 58 and 1925 ± 60 m.y. were obtained on contact biotite and chilled whole-rock of one of the dikes (analysis 66-132 and 70-97 of GSC papers 67-2 and 71-2). Thus the Otish Mountains Group is certainly older than 1787 m.y. and possibly older than 1960 m.y. Frarey and Roscoe (1970) suggested the Otish Mountains Group may be correlative with the Huronian Supergroup, which has an age of 2300 m.y. (Fairbairn *et al.* 1969).

## Paleomagnetism

Seventy-five cores were drilled and oriented (by solar compass) at the 10 sites whose location is indicated on Fig. 1. Two specimens were cut from each core and an optimum A.F. cleaning range for each site was chosen by step-cleaning 2 specimens per site. Most of the step-cleaned specimens exhibited good end-points (Table 3) and on a basis of these results, all specimens from each site were cleaned in at least two field intensities within the range suggested by step-cleaning. Data from cores which, after cleaning at optimum fields, exhibited between-specimen directions that differed by more than 25° ($\theta > 25°$) were not included in the final statistics.

[1]Decay constant $1.39 \times 10^{-11}\text{yr}^{-1}$.

TABLE 2. Magnetic properties of the Otish Gabbro

| | n.r.m. | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Site | *M* | *N* | *n* | *D*° | *I*° | k | δ° | *A.F.* | *D*° | *I*° | k | δ° | $\delta^{\circ}_{R}$ | *A* |
| 53 | 557 | 7 | 3 | 285 | 79 | 597 | 3.3 | 250 | 253 | 67 | 157 | 6.5 | 3.7 | M |
| 54 | 2137 | 8 | 5 | 292 | 84 | 16 | 20 | 250 | 271 | 71 | 18 | 19 | 8.5 | M |
| 55 | 2850 | 9 | 7 | 298 | 81 | 25 | 16 | 150 | 277 | 75 | 75 | 9.4 | 3.5 | F |
| 56 | 23 | 9 | 7 | 220 | 85 | 115 | 7.5 | 150 | 217 | 78 | 85 | 8.8 | 3.3 | I |
| 57 | 409 | 8 | 6 | 356 | 82 | 139 | 6.9 | 200 | 263 | 78 | 47 | 12 | 4.8 | I |
| 58 | 1600 | 6 | 6 | | RANDOM | | | 150 | 247 | 60 | 43 | 12 | 5.1 | M |
| 59 | 4470 | 6 | 5 | | RANDOM | | | 150 | 205 | 78 | 206 | 5.6 | 2.5 | F-M |
| 60 | 2490 | 7 | 5 | 309 | 81 | 52 | 8.7 | 200 | 277 | 63 | 201 | 5.7 | 2.6 | F |
| 61 | 324 | 6 | 5 | 158 | 88 | 46 | 12 | 200 | 218 | 69 | 220 | 5.5 | 2.4 | M |
| 62 | 76 | 9 | 7 | | RANDOM | | | 250 | 244 | 78 | 60 | 10 | 4.0 | M-I |
| All cores | | 75 | 56 | 253 | 87 | 6 | 33 | | 249 | 74 | 36 | 14 | 1.8 | |
| All sites | | 10 | 10 | 242 | 87 | 29 | 15 | | 250 | 73 | 64 | 10 | 3.2 | |

Mean pole position from 10 site poles 106.6° W, 34.8° N, $K = 24$, $\alpha^{\circ}_{95} = 9.0°$, $\delta^{\circ}_{R} = 16$

$M$ = mean site intensity of n.r.m. $\times 10^{-6}$ c.g.s./cm$^3$, $N$ = number of samples collected at the site, $n$ = number of samples left after eliminating those cores which are inhomogenously magnetized (θ greater than 25°). $D$ = declination, $I$ = inclination, δ° angular standard deviation, A.F. = peak alternating field in oersteds, $k$ = Fisher's estimate of precision, $\alpha^{\circ}_{95}$ = half angle of the cone of confidence at $P = 0.95$, $\delta^{\circ}_{R}$ = standard error of the mean.

$A$ = site mean alteration (including uralitization) as estimated by examining one thin section from each of the 75 cores: $F$ = fresh, $M$ = moderately altered, $I$ = intensely altered.

TABLE 3. Directional changes during step demagnetization of one specimen per site

| Site | 53 | | 54 | | 55 | | 56 | | 57 | |
|---|---|---|---|---|---|---|---|---|---|---|
| Field | *D* | *I* | *D* | *I* | *D* | *I* | *D* | *I* | *D* | *I* |
| 0 | 291 | 80 | 282 | 72 | 58 | 80 | 243 | 81 | 40 | 72 |
| 50 | 278 | 75 | 275 | 71 | 275 | 84 | 202 | 70 | 230 | 74 |
| 100 | 272 | 68 | 268 | 68 | 284 | 71 | 218 | 70 | 228 | 72 |
| 150 | 235 | 66 | 287 | 69 | 301 | 72 | 199 | 69 | 225 | 65 |
| 200 | 255 | 62 | 258 | 63 | 239 | 90 | 175 | 73 | 203 | 67 |
| 250 | 252 | 55 | 266 | 61 | 12 | 45 | 306 | 70 | 275 | 70 |
| 350 | 250 | 50 | 254 | 56 | 260 | 25 | 302 | 71 | 13 | 55 |
| 500 | 210 | 32 | 212 | 62 | 271 | 14 | 3 | 72 | 122 | 57 |
| 800 | 221 | −16 | 224 | 34 | 107 | −54 | 60 | −9 | 267 | 26 |
| | 58 | | 59 | | 60 | | 61 | | 62 | |
| 0 | 278 | 81 | 195 | 24 | 270 | 80 | 234 | 78 | 27 | 36 |
| 50 | 262 | 74 | 182 | 57 | 269 | 73 | 226 | 75 | 24 | 38 |
| 100 | 257 | 66 | 180 | 74 | 276 | 68 | 226 | 75 | 358 | 64 |
| 150 | 234 | 54 | 173 | 84 | 269 | 61 | 227 | 74 | 310 | 70 |
| 200 | 231 | 47 | 249 | 71 | 281 | 57 | 228 | 73 | 283 | 69 |
| 250 | 286 | 84 | 278 | 46 | 277 | 55 | 220 | 74 | 288 | 73 |
| 350 | 215 | 73 | 106 | 31 | 283 | 61 | 223 | 66 | 167 | 84 |
| 500 | 121 | 35 | 301 | 25 | 236 | 55 | 218 | 72 | 36 | 52 |
| 800 | 106 | −5 | 200 | 14 | 254 | 26 | 191 | 37 | 67 | 18 |

The directional changes of the test specimens of sites 53, 54, 60, 61 above, appear to indicate a continuous shift away from the present earth's field direction at higher cleaning fields. This was not true of other test specimens from these sites (data from which were also used in choosing an optimum site cleaning field), nor do the test specimens, as a group, show this systematic shift at higher fields.

One core from each of 5 sites exhibited n.r.m. intensities that were much greater than those of the remaining cores of these sites: the highest initial intensity, $18 \times 10^{-3}$ c.g.s. cm$^{-3}$. Some of these cores may have been exposed to lightning strikes, as most sites were located on prominent ridges. Two cores exhibited n.r.m. intensities that were notably weaker ($10^2$) than the other cores of their sites. If data for these 7 cores are excluded, the within-site variation in initial intensity is reasonably small (less than a factor of 10) and the arithmetic means pro-

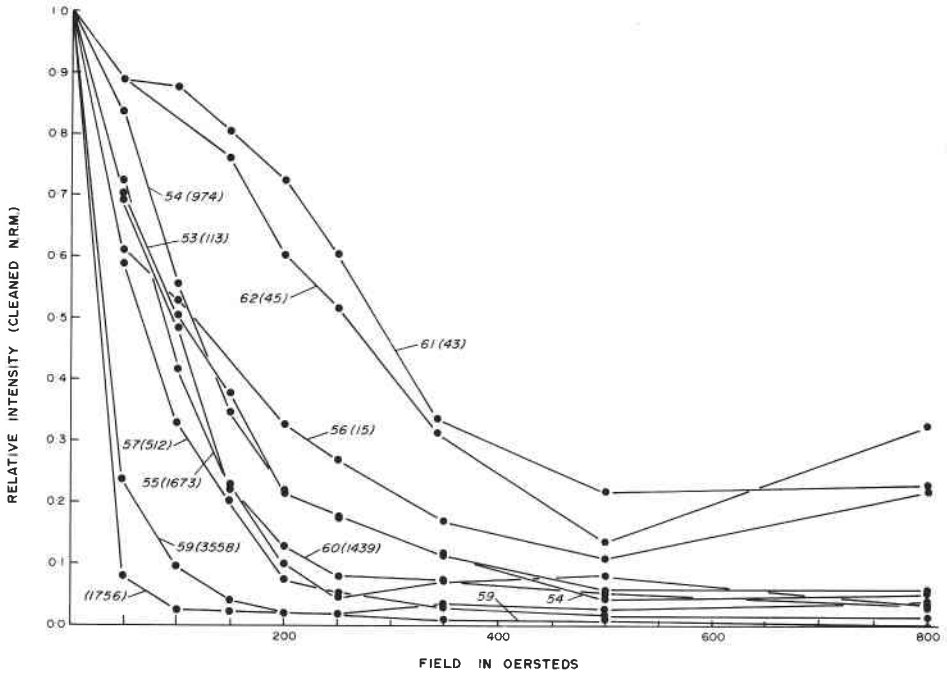



FIG. 2. Normalized A.F. demagnetization curves of one specimen from each site. The number beside each curve identifies the site from which the specimen was taken and gives the n.r.m. intensity (bracketed) in e.m.u. $\times 10^{-6}$.

vide a good comparison of between-site initial intensities. The data of Table 2 indicate a large variation in between-site intensity ($2 \times 10^2$), which is notably greater than that generally encountered in similar rocks, for example, the Franklin diabases (Fahrig *et al.* 1971). This is also clear from the demagnetization curves of individual specimens (Fig. 2). The coercivity of the n.r.m. (as indicated by the median destructive alternating field) is higher for weakly magnetic specimens (see specimens from sites 61 and 62 of Fig. 2). Since several of the sites are from the same sill, the variation in initial intensity must be due either to primary differences in the igneous mineralogy or to differences due to the late magmatic (deuteric) alteration referred to above.

One thin-section from each of the 75 cores was examined and the degree of alteration of the silicates was estimated (Table 2). It is clear that in all but site 54 a correlation exists between the initial intensity of magnetization and the extent of alteration. It is assumed then that the wide variation in initial intensity is due to this low temperature alteration. The alteration has led to a decrease in intensity, and as this alteration varies widely the initial intensity does the same. If this alteration, as suggested above, was related to the cooling of the gabbro, any chemo-remanent magnetization component should have the same direction as the thermo-remanent magnetization.

The optimum cleaning fields for the 10 Otish Gabbro sites ranged from 150 to 250 Oe, and most of the test specimens exhibited large random directional changes after cleaning in peak alternating fields above 250 Oe. In most cases within-site grouping improved noticeably in response to the A.F. treatment, as did the between-site grouping (Fig. 3 and Table 2). Most of the site-mean directions shifted away from the recent earth's field direction with cleaning, and the mean inclination of the magnetization of the gabbro decreased by about 15°. As the only alteration of these rocks was an early deuteric uralitization (a suggestion supported by the argeement among sites from the same sill, which have undergone variable amounts of alteration), the mean direction probably represents the earth's field

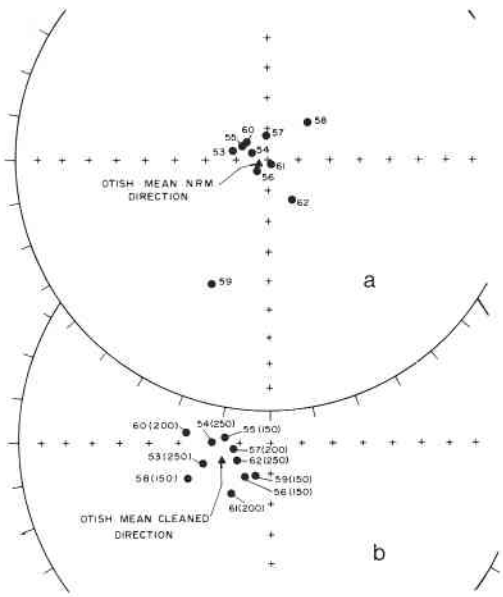



FIG. 3. Equal-area projection of mean site directions, all of which are downward pointing; n.r.m. directions (a), cleaned directions (b). Numbers near the dots identify the sites, bracketed numbers in (b) indicate cleaning field strength in oersteds.

direction relative to this area of crust at the time of intrusion. Figure 1 lists the dip of the contact between the sills and underlying sedimentary rocks at most of the sites. These contacts dip only a few degrees and as the sills may have had small primary south dips no correction has been made for the observed dips. There is no may to demonstrate conclusively that the sampling adequately averages the paleosecular field variation at the time of Otish Gabbro intrusion. We feel, however, that the two major sills, a dike feeder to a third major sill, as well as various feeder dikes and subsidiary sills were probably intruded over a suitably long period of time.

## Discussion

The paleomagnetism of the Otish Gabbro was studied in the hope that data of this type would contribute to our understanding of the correlation of the Otish Mountains Group and to our knowledge of the North American Precambrian polar-wandering curve. It had been suggested on stratigraphic grounds (Bergeron 1957; Chown and Caty 1973) that the Otish Mountains and Mistassini Groups are correlatives and that in a general way these groups correlate with rocks of the Labrador Trough and Belcher Islands. On the other hand, Frarey and Roscoe (1970) suggested that the Otish Mountains Group is similar lithologically and in age to the Huronian Supergroup. This would make the Otish about 2300 m.y. old—significantly older than strata of the Mistassini Group, or those of the Labrador Trough and Belcher Islands.

Paleomagnetic data for rocks comparable in age to the Otish Mountains Group and which either lie undisturbed on, or are intrusive into, the rocks of the Superior Orogen are rare. The Nipissing diabase has a minimum age between 2100 and 2200 m.y. (Lowden *et al.* 1963; van Schmus 1965; Fairbairn *et al.* 1969). In the Cobalt district it has not been significantly deformed, is reversely magnetized, and yields a paleopole of 088.1° W, 19.4° S (Symons 1970), and Nipissing sheets near Elliot Lake have a similar pole position (Symons 1971). The combined Nipissing pole position therefore lies about 55° south of that obtained for the Otish Gabbro. The Sudbury norite is approximately 1700 m.y. old[2] (Fairbairn *et al.* 1968) and the pole from the north range lies about 20° north (Fig. 4) of the Otish pole (Sopher 1963). These data suggest that either the Otish Mountains Group (and its contained Otish Gabbro) is younger than the Nipissing diabase (and the Huronian Supergroup which it intrudes), or the Otish Gabbro was formed a considerable time after the deposition of the sedimentary strata (Fig. 4).

Paleomagnetic results indicate that the latitude of the Otish Gabbro was 60° N when these rocks acquired their stable magnetization, presumably at the time of intrusion. The upper formation of the Otish Mountains Group contains stromatolites, which may argue against deposition at high latitude. However, the Otish Gabbro may have been intruded (or less likely, remagnetized) long after the deposition of the sedimentary strata.

[2]Gibbins and McNutt have published an age of 2000 m.y. for the irruptive. It is difficult on the basis of only an abstract to properly evaluate this age determination. If it is correct, the Otish pole is more closely tied between the 2200 m.y. Nipissing and the 2000 m.y. old Sudbury pole.

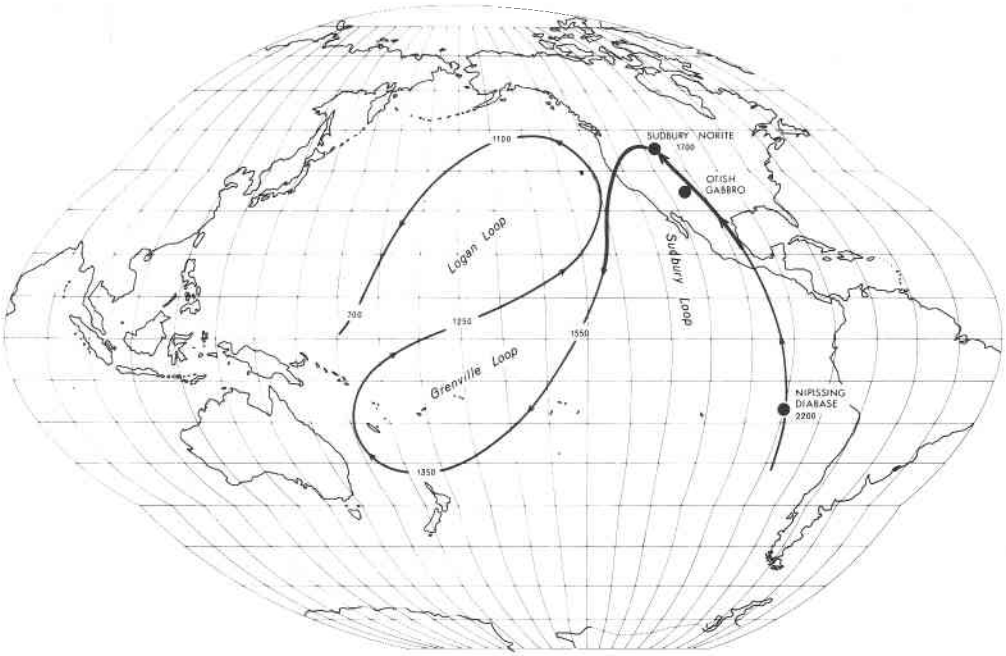



FIG. 4. The pole position of the Otish Gabbro and those of rocks older (Nipissing 2200 m.y.) and younger (Sudbury 1700 m.y.) from the Superior Orogen. The Precambrian polar wandering curve for the Superior Province between 1500 and 1700 m.y. is taken from Robertson and Fahrig (1971) and Roy and Fahrig (1973). Data used in the construction of this section are not referenced here because of the brevity of the present study.

The pole for the Sudbury norite is that obtained from the North Range assuming post-magnetization tilting of 10° (Sopher 1963). It is possible that because of greater tilting the pole should be somewhat south of the one shown here. An attempt to examine this tilt problem by study of the Dubawnt Group, N.W.T. (Park *et al.* 1973) appears to be of doubtful value. The writers have equated the ages of the Sudbury norite and Dubawnt Groups without reference to the different decay constants used in the Rb/Sr age determinations. If decay constants are equated the Dubawnt is probably at least 125 m.y. older than the Sudbury. Furthermore it may be risky to assume that no movement has taken place between the Churchill and Superior Orogens in the past 1800 m.y.

## Acknowledgments

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BERGERON, R. 1957. Late Precambrian rocks of the North Shore of the St. Lawrence River and of the Mistassini and Otish Mountains Areas, Quebec. *In*: The Proterozoic in Canada, J. E. Gill (*Ed.*), Univ. Toronto Press, Toronto, Ontario, pp. 124–132.

CHOWN, E. H. 1968. Geology of the Conflans Lake area. Quebec Dep. Natur. Resour. Unpubl. ms., open file, Quebec, P.Q.

——— 1969. Geology of the Pariseau Lake area. Quebec Dep. Natur. Resour. Unpubl. ms., open file, Quebec, P.Q.

——— 1970. Geology of the Pluto Lake area, Quebec Dep. Natur. Resour. Prel. Rep. No. 584.

CHOWN, E. H. and CATY, J. L. 1973. Stratigraphy, petrography, and paleocurrent analysis of the Aphebian clastic formations of the Mistassini–Otish basin. *In*: Huronian sedimentation, G. M. Young (*Ed.*), Geol. Ass. Can. Spec. Publ. 12.

FAHRIG, W. F., IRVING, E., and JACKSON, G. D. 1971. Paleomagnetism of the Franklin diabases. Can. J. Earth Sci., **8**, pp. 455–467.

FAIRBAIRN, H. W., FAURE, G., PINSON, W. H., and HURLEY, P. M. 1968. Rb–Sr whole-rock age of the Sudbury lopolith and basin sediments. Can. J. Earth Sci., **5**, pp. 707–714.

FAIRBAIRN, H. W., HURLEY, P. M., CARD, K. D., and KNIGHT, C. J. 1969. Correlation of radiometric ages of Nipissing diabase and Huronian metasediments with Proterozoic orogenic events in Ontario. Can. J. Earth Sci., **6**, pp. 489–497.

FRAREY, M. J. and ROSCOE, S. M. 1970. The Huronian Supergroup north of Lake Huron. *In*: Symposium on basins and geosynclines of the Canadian Shield, A. J. Baer (*Ed.*), Geol. Surv. Can., Pap. 70-40, pp. 143–159.

FRYER, B. J. 1972. Age determinations in the Circum–Ungava Geosyncline and the evolution of Precambrian banded iron-formations. Can. J. Earth Sci. **9**, pp. 652–663.

GIBBINS, W. A. and MCNUTT, R. H. 1972. Rb/Sr Studies of the Sudbury Irruptive. Geol. Soc. Am. Abstr., **4**, p. 517.

LOWDEN, J. A., STOCKWELL, C. H., TIPPER, H. W., and WANLESS, R. K. 1963. Age determinations and geological studies. Geol. Surv. Can., Pap. 62-17, 140 p.

MACDONALD, G. H. and KATSURA, T. 1964. Chemical composition of Hawaiian lavas. J. Petrol., **5**, pp. 82–133.

NEILSON, J. M. 1966. The Takwa River area. Quebec Dep. Natur. Resour. Geol. Rep. 124, 53 p.

PARK, J. K., IRVING, E., and DONALDSON, J. A. 1973. Paleomagnetism of the Precambrian Dubawnt Group. Bull. Geol. Soc. Am., **85**, pp. 859–870.

ROBERTSON, W. A. and FAHRIG, W. F. 1971. The great Logan paleomagnetic loop—the Polar wandering path from Canadian Shield rocks during the Neohelikian Era. Can. J. Earth Sci., **8**, pp. 1355–1372.

ROY, J. L. and FAHRIG, W. F. 1973. The paleomagnetism of Seal and Croteau rocks from the Grenville Front, Labrador; Polar wandering and tectonic implications. Can. J. Earth Sci., **10**, pp. 1279–1301.

SOPHER, S. R. 1963. Paleomagnetic study of the Sudbury Irruptive Geol. Surv. Can., Bull. 90, 36 p.

SYMONS, D. T. A. 1970. Paleomagnetism of the Nipissing diabase, Cobalt area, Ontario. Can. J. Earth Sci., **7**, pp. 86–90.

——— 1971. Paleomagnetic study of the Nipissing diabase, Blind River – Elliot Lake area, Ontario. *In*: Geol. Surv. Can., Pap. 70-63, pp. 19–30.

VAN SCHMUS, R. 1965. The geochronology of the Blind River – Bruce Mines area, Ontario, Canada. J. Geol., **73**, pp. 755–780.

WANLESS, R. K., STEVENS, R. D., LACHANCE, G. R., and RIMSAITE, R. Y. H. 1965. Age determinations and geological studies. Part 1 – isotope ages, Rep. 5. Geol. Surv. Can., Pap. 64-17, pt. 1, 126 p.